

JEE MAIN REVISION MASTER: MOVING CHARGES AND MAGNETISM 2026 4 QUESTIONS 6.67% WEIGHTAGE ↓19.93%

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 48 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Moving Charges And Magnetism - Core Application	48 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Moving Charges And Magnetism - Core Application (Very High Yield — 48 PYQs)

Core Principle: Key problem pattern in Kcet Moving Charges And Magnetism.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this cha..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "Two identical circular current loops carrying equal currents are placed with their axes inclined at 45° to each other as shown in t..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "Biot-Savart law indicates that an electron moving with a velocity $\vec{v}$ produces a magnetic field $\vec{B}$ around it such that..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A point charge is placed in a moving train. A passenger A sitting in the train and person B on the ground observe the fields due to this charge. Then

- (A) observes both electric and magnetic fields
- (B) observes both electric and magnetic fields
- (C) observes only magnetic field
- (D) observes only electric field

Correct Answer: None

Solution Analysis: The correct option is: Option B: B observes both electric and magnetic fields The point charge is placed in the moving train. For passenger A sitting inside the train, the charge is at rest. A stationary charge produces only an electric field around it. So A observes only electric field, not magnetic field. For person B standing on the ground, the train is moving. So the charge is also moving with the train. A moving charge is equivalent to an electric current, and current produces a magnetic field. Since the charge itself also has electric field, B observes both electric and magnetic